Study of different type of Network cable

Exp no 2

a Understand different types of network cable.

Different type of cable used in networking are

1. Unshielded Twisted Pair (UTP) cable
2. Shielded Twisted Pair (STP) cable
3. Coaxial Cable
4. Fibre optic Cable

cable type	category	Bandwidth	Advantage / Disadvantage	Application	Image
UTP	category 3	10 bps	Advantage: cheaper in cost	10 Base T Ethernet	
	category 5	upto 100 Mbps	Easy to install as they have a smaller overall dimeter	Fast Ethernet	
	category 5e	1 Gbps			
STP	Category 6, 6a	10 Gbps	Advantage: Shielded; Faster than UTP; less Susceptible to noise and interference	Gigabit Ethernet; 10 G	

SSTP	category 7	10 Gbps	Disadvantages • Expensive • Creates installation
coaxial	RG-6 RG-59 RG-11	10-100 MBps	• High bandwidth • Immune to interference • low loss bandwidth Disadvantage • Limited distance • Cost
fibre optical	single mode	100Gbps	Advantage • High speed • High bandwidth • Long distance Disadvantage • Expensive • Requires skilled installers

1) Make your own Ethernet cross-over cable / Straight cable

Tools and parts needed:

- Ethernet cabling. CAT5e is certified for gigabit support, but ea. CAT5 cabling work well, just over short distance
- A crimping tool. This is an all-in-one network tool shaped to push down the pins in the plug and strip and cut the shielding off the cable
- Two RJ45 plugs
- Optional two plug shields

Result:
Thus different type of cable network are learned successfully